Griffiths Chapter 6, Section 6.3.3

Higher-Order Diagrams & the Road to Renormalization

1. Where We Are in the Story

By the time you reach Section 6.3.3, you've built up the complete lowest-order machinery of the Feynman calculus in the ABC toy theory. You know how to:

(a) Write down the amplitude M for a process by drawing the relevant Feynman diagram and applying the six Feynman rules (label momenta, write vertex factors -ig, write propagators $i/(q^2 - m^2c^2)$, enforce 4-momentum conservation at each vertex, integrate over internal momenta, cancel the overall delta function and multiply by i).

(b) Plug $|M|^2$ into the Golden Rule to get a decay rate (Eq. 6.35 for two-body decays) or a differential cross section (Eq. 6.47 for two-body scattering in the CM frame).

(c) Compute actual numbers: the A lifetime (Section 6.3.1), and the A+A $\rightarrow$ B+B differential cross section (Section 6.3.2).

All of this used only **tree-level** (lowest-order) diagrams — diagrams with no closed loops. Section 6.3.3 asks: what happens when we go beyond tree level?

2. Tree Level vs. Loop Diagrams

A tree-level diagram is one you can draw without any internal lines forming a closed loop. It is called 'tree level' because, like a tree, it branches but never reconnects with itself. Every vertex brings one power of the coupling constant g, so the order of a diagram is determined by counting vertices.

> **Physical intuition:** A tree-level diagram describes the simplest quantum-mechanical process — a single, direct interaction path. Loop diagrams represent *quantum corrections*: the intermediate particles can briefly fluctuate into virtual particle-antiparticle pairs or exchange additional virtual quanta before the process completes. These are the quantum field theory analog of perturbative corrections in ordinary quantum mechanics.

2.1 Counting the Order

For A+A $\rightarrow$ B+B, the lowest-order diagrams have **two vertices**, so the amplitude goes as g^2. The next corrections have **four vertices**, giving amplitudes proportional to g^4. Since the coupling constant g is (presumably) small, each additional pair of vertices suppresses the contribution — this is why perturbation theory works, and why tree-level results are often a good first approximation.

3. The Zoo of One-Loop Diagrams

Griffiths catalogs the one-loop (order g^4) corrections to A+A $\rightarrow$ B+B. There are **eight diagrams** (plus eight more with the external B lines 'twisted'). They fall into three categories:

Type	Count	What Happens Physically
Self-energy diagrams	5	One of the propagator lines 'sprouts' a loop — a virtual particle-antiparticle bubble appears on an internal or external line. This modifies the effective mass of the particle carrying that line.
Vertex corrections	2	A vertex 'opens up' into a triangle — the single interaction point is replaced by a loop connecting three internal lines. This modifies the effective coupling constant.
Box diagram	1	A completely new topology: a closed rectangular loop connects the four external lines. This represents a genuinely new scattering pathway with no tree-level analog.

Why these three types matter: Each type of loop correction maps onto a specific physical effect. Self-energy loops shift particle masses. Vertex corrections shift coupling strengths. Box diagrams contribute new scattering channels. When you encounter QED in Chapter 7, these same three categories reappear — vacuum polarization (self-energy on the photon line), electron self-energy, vertex corrections, and box diagrams.

4. The Bubble Diagram: A Detailed Calculation

Griffiths works through one specific loop diagram in detail: the 'bubble' on the virtual C line in A+A $\rightarrow$ B+B. This is the simplest self-energy correction, and it perfectly illustrates both the power and the problem of loop calculations.

4.1 Setting Up the Integral

The bubble diagram has **four vertices** (two from the original tree diagram, plus two where the bubble attaches), **three internal lines** (the original C propagator is replaced by two propagators forming the loop, plus the C line connecting to the other vertex), and **one undetermined loop momentum** q.

Applying the Feynman rules systematically — writing down vertex factors, propagators, delta functions, and integrating over internal momenta — Griffiths reduces everything to a single remaining integral over the loop momentum q:

$$M_{bubble} \sim g^4 / [(p_1-p_3)^2 - m_C^2c^2]^2 \times \int d^4q / [(q^2 - m_A^2c^2)(q'^2 - m_B^2c^2)]$$

where $q' = (p_1 - p_3) - q$ is fixed by momentum conservation at the bubble vertices. The key point is that after all the delta functions are used up, one four-momentum integral remains —

the loop momentum q is *not* determined by conservation laws. You must integrate over all possible values of q. This is fundamentally different from tree-level diagrams, where all internal momenta are completely fixed by the external kinematics.

Key insight: In a tree diagram, every internal momentum is determined by energy-momentum conservation at the vertices — no free integrations remain. In a loop diagram, there is always at least one unconstrained momentum that must be integrated over. Each loop adds one such integration. This is the mathematical origin of both the richness and the trouble with loop diagrams.

5. The Divergence: Where Things Blow Up

Here is the central drama of Section 6.3.3. The loop integral over d^4q diverges. Griffiths explains why with a simple power-counting argument:

The four-dimensional volume element can be written as $d^4q \sim q^3\, dq$ (times angular factors), just as $d^3x = r^2\, dr\, d\Omega$ in 3D. At large q, each propagator in the loop goes as $1/q^2$, so two propagators give $1/q^4$. The integrand at large q therefore behaves as $q^3/q^4 = 1/q$. The integral becomes:

$$\int (1/q)\, dq = \ln(q) \,|\, \to \infty$$

This is a **logarithmic divergence** — the integral blows up, but only as slowly as possible (logarithmically rather than as a power of the cutoff). This is actually the mildest kind of infinity you can get, and that mildness turns out to be crucial for the success of renormalization.

> **Power counting rule (useful for all of QFT):** For a loop integral in 4 spacetime dimensions, count the powers of momentum in the numerator (from d^4q and any momentum factors in vertices) versus the denominator (from propagators, each giving q^2). If [numerator powers] - [denominator powers] ≥ 0, the integral diverges. Equal powers $\to$ logarithmic divergence. More on top $\to$ power-law divergence.

6. Regularization: Taming the Infinity

To make mathematical progress, you need to *regularize* the integral — impose some procedure that renders it finite so you can manipulate it, with the intention of taking a limit at the end. Griffiths introduces a **Pauli-Villars** style cutoff: insert a factor $(q^2 - M^2c^2)^{-1}$ under the integral, where M is a very large 'cutoff mass.' This suppresses the high-momentum contributions and makes the integral finite.

After regularization, the integral separates into two pieces: a **finite part** that is independent of the cutoff M and depends on the physical momenta, plus a **divergent part** that depends on ln(M) and blows up as $M \to \infty$.

> The physical picture: the cutoff M is a confession of ignorance. We don't really know what happens at arbitrarily high energies (arbitrarily short distances). Perhaps the particles have internal structure, or spacetime itself is discrete, or new physics appears. The cutoff parametrizes that ignorance.

7. The Miracle of Renormalization

Now comes what Griffiths calls 'a miraculous thing.' When you carefully examine where the divergent (M-dependent) terms end up in the final answer, they *all* appear as additive corrections to the masses and the coupling constant. Not scattered randomly through the result — specifically attached to m_A, m_B, m_C, and g.

This means the divergences don't actually produce new, unphysical effects. They just shift the values of quantities (masses and couplings) that we were going to measure experimentally anyway. The procedure is:

Step 1: Recognize that the 'bare' parameters m and g appearing in the original Lagrangian are *not* the physically measured quantities. They are unmeasurable mathematical inputs.

Step 2: Define the **physical** (renormalized) mass and coupling as the bare value plus the (infinite) correction: $m_{phys} = m + \delta m$, and $g_{phys} = g + \delta g$.

Step 3: Rewrite all results in terms of m_{phys} and g_{phys}. The infinities cancel, and what remains are finite, calculable corrections that depend on the momenta of the particles involved.

The bare masses and couplings are individually infinite (as $M \to \infty$), but that's okay — we never measure them directly. All we ever measure in the lab are the physical values, which are manifestly finite. The infinite bare quantities contain 'compensating infinities' that exactly cancel the loop divergences.

8. Running Masses and Couplings

Beyond the divergent pieces, the loop diagrams also produce **finite corrections** that depend on the four-momenta flowing through the diagram. This has a remarkable physical consequence: the effective mass and coupling constant *depend on the energy scale* at which you probe them. These are called **running** masses and **running** coupling constants.

At low energies the running is typically very slight and can be ignored. But it has real, observable consequences. Griffiths flags two that you'll see in later chapters: the **Lamb shift** in QED (a tiny energy splitting in hydrogen that comes from vacuum polarization — the photon propagator getting 'dressed' by virtual electron-positron loops), and **asymptotic freedom** in QCD (the strong coupling constant gets weaker at high energies, which is why quarks behave almost freely inside protons at short distances).

Preview for QED (Chapter 7): The bubble diagram you studied here in ABC theory is the direct ancestor of *vacuum polarization* — the virtual photon momentarily splitting into an e^+e^- pair. The mathematics is more involved (spinors, gamma matrices), but the conceptual structure is identical: a loop integral diverges, you regularize it, and the infinities get absorbed into a redefinition of the electric charge $e \to e_R$. The finite remainder gives the running of the fine structure constant $\alpha(q^2)$.

9. Summary: What to Take Away

1. **Tree diagrams** have no loops. All internal momenta are fixed by conservation laws. They give the leading-order (lowest power of g) contribution to any process.

2. **Loop diagrams** introduce unconstrained momentum integrals — one free integration per loop. They represent quantum corrections (virtual fluctuations) to the tree-level result.

3. **Higher-order diagrams** for A+A → B+B at one-loop level come in three flavors: self-energy corrections (5 diagrams), vertex corrections (2 diagrams), and box diagrams (1 diagram). Plus 8 more from 'twisting' the external B lines.

4. **Loop integrals diverge.** Power counting (comparing powers of momentum in numerator vs. denominator) tells you how badly. The bubble diagram in ABC theory is logarithmically divergent.

5. **Regularization** is a mathematical device to make divergent integrals temporarily finite (e.g., introducing a large cutoff mass M). It parametrizes our ignorance of very-high-energy physics.

6. **Renormalization** is the key physical insight: the divergent parts of loop corrections are not random garbage — they are precisely corrections to masses and coupling constants. Absorbing them into 'physical' (renormalized) values of m and g removes all infinities.

7. **Running couplings:** The finite leftover from loop corrections makes the effective mass and coupling depend on energy. This is physically real and leads to the Lamb shift (QED) and asymptotic freedom (QCD).

8. **A theory is renormalizable** if *all* divergences can be absorbed into a finite number of physical parameters. The ABC toy theory is renormalizable, as are QED and QCD. This is not guaranteed for every theory — it's a deep structural property.

10. Practice & Self-Test

Try these to check your understanding. Hints follow each question.

Problem 1. Consider the process A+B → A+B in the ABC theory. At lowest order, there are two contributing diagrams (Griffiths Problem 6.15a). Draw them. How many vertices does each have? What order in g is the total amplitude?

Hint: Each diagram has two vertices (one A-B-C vertex at each end), connected by a single internal C line. One has a C exchanged in the 't-channel' and the other in the 'u-channel.' Both are order g^2, so $M \sim g^2$.

Problem 2. For the bubble diagram on the C line, Griffiths gets a loop integral that goes as ∫ d■q / q■ at large q. Suppose instead we were in a 6-dimensional spacetime. Would the divergence be better or worse?

Hint: In d dimensions, $d^4q \to d^dq \sim q^{d-1}dq$. With two propagators ($\sim 1/q^4$), the integrand goes as q^{d-5}. For d=4: q^{-1} (log divergent). For d=6: q^1 (quadratically divergent — much worse!). Higher dimensions make divergences more severe. This is why renormalizability depends on spacetime dimension.

Problem 3. Why can't the A+A → B+B process have a one-loop diagram with *three* vertices?

Hint: Each vertex in ABC theory connects exactly one A, one B, and one C line. The number of vertices must be even for A+A → B+B (or any process that's allowed at all), because each vertex changes the particle count by fixed amounts. Three vertices would give an amplitude ~ g^3, but there's no way to connect 2 external A's and 2 external B's with an odd number of ABC vertices while conserving particle type at each vertex.

Problem 4. Griffiths says 'the physical masses and couplings are not the m's and g's that appeared in the original Feynman rules.' In your own words, explain why this makes physical sense.

Hint: Think of it by analogy with electrostatics: the 'bare' charge of an electron sitting in a dielectric medium is not what you measure from far away — the medium screens it. Similarly, a 'bare' particle in QFT is surrounded by a cloud of virtual fluctuations that modify its effective mass and charge. What we measure in the lab already includes all of these dressing effects. The bare parameters are just mathematical starting points.

This summary covers Griffiths §6.3.3 with context from Thomson (Modern Particle Physics, Ch. 10.5) and Lancaster & Blundell (QFT for the Gifted Amateur, Chs. 32–34). The conceptual framework — tree vs. loop, divergence, regularization, renormalization, running couplings — is the foundation for everything in Chapters 7–10.